

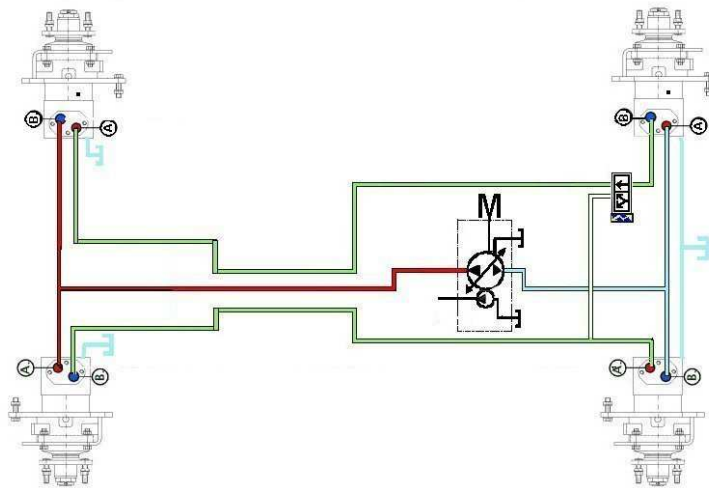


Avant Drive Release-, and the Anti Slip Valve

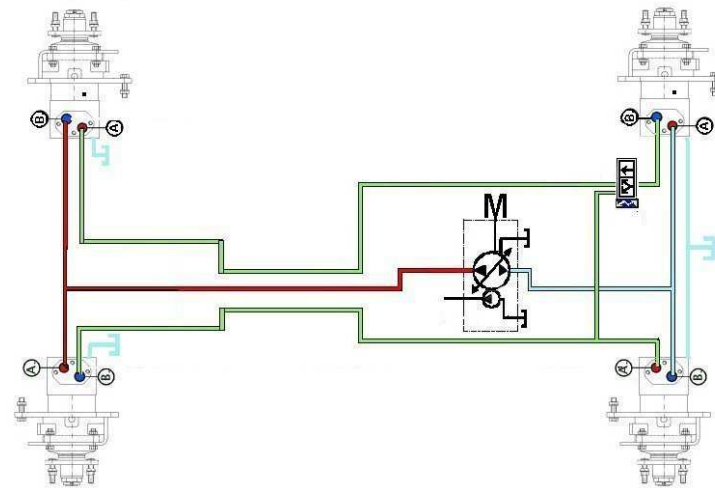


Avant Drive Release Valve

LOCK ON : Valve closed or it does not exist. Solenoid valve inactive. The free flow of hydraulic oil, between the drive motor pairs at the right and the left side has been hindered. Mobility and traction is sufficient in most ground environments.



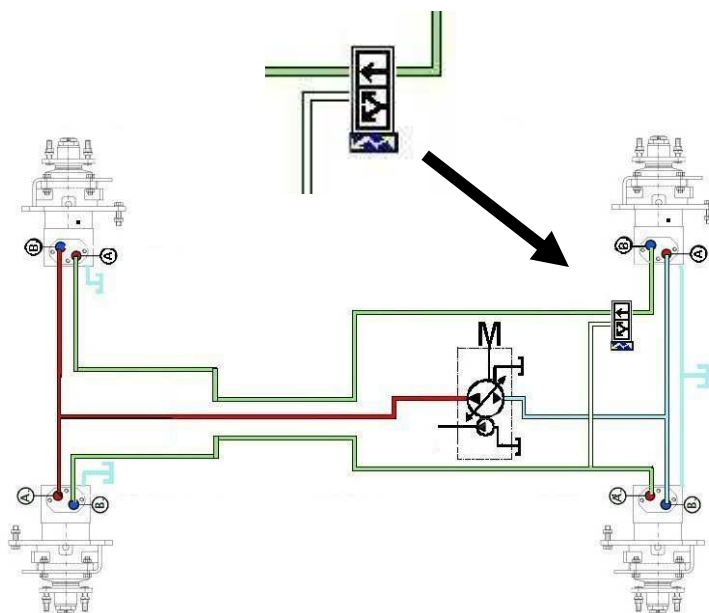
LOCK OFF: Valve open. The oil flow, allowed between all drive motors. The oil flow is free to rotate the motors to the selected direction, now the tyres leaving a minimal amount of trackmarks when driving and steering the loader at any surfaces.



Avant Drive Release Valve

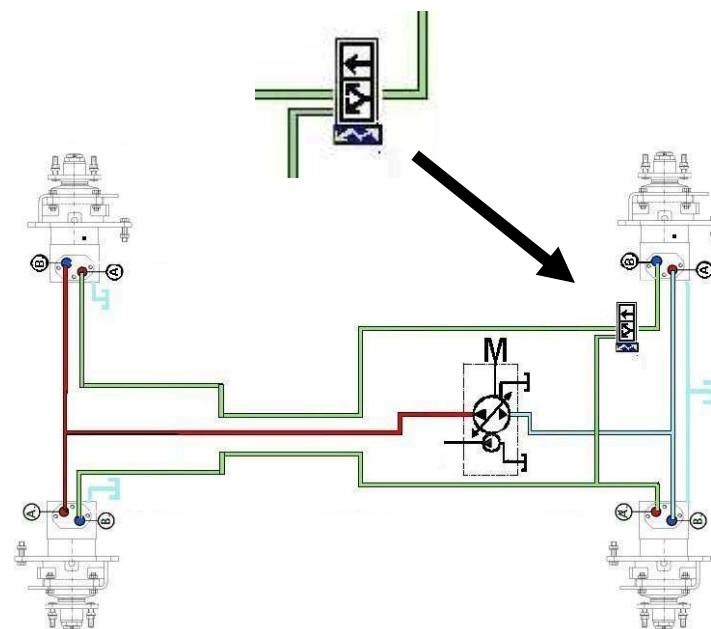
LOCK ON :

The drive release valve at the normal state, not activated



LOCK OFF:

The drive release valve activated





Avant Drive Release Valve

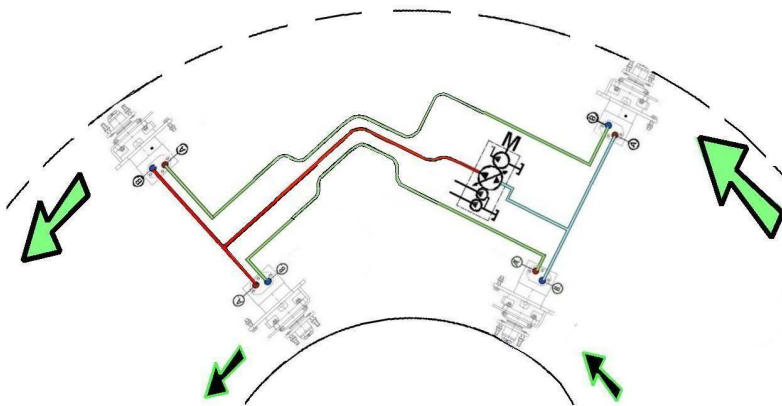


Avant Drive Release Valve

The drive release valve becomes necessary :

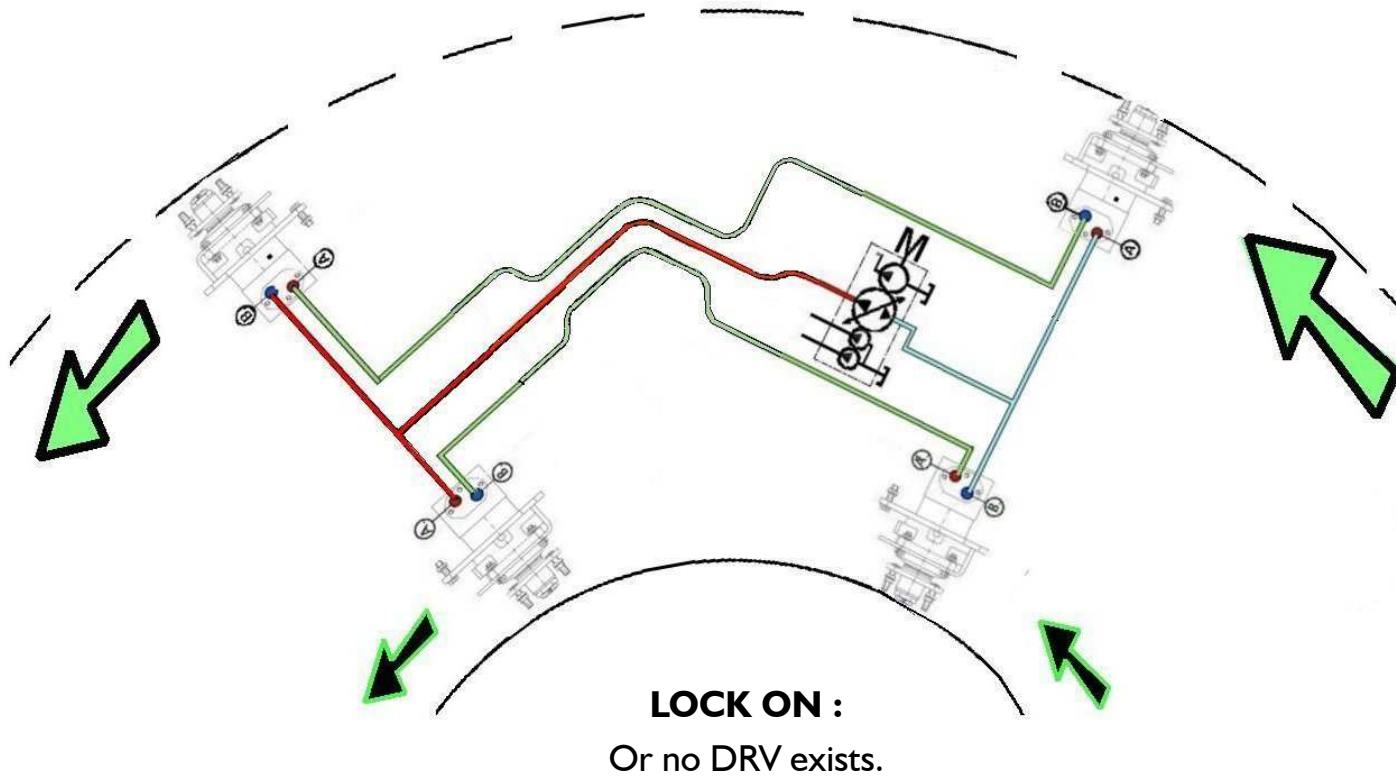
A.) When sharp turns and bends will be driven repeatedly, the wheel motors at outer radius need more oil volume to rotate evenly compared to the ones at the inner radius. With the help of this valve, most of the undesired trail marks to sensitive surfaces can be minimised.

B.) When steering a stand still Avant loader, or driven with a minimal speed the activated valve makes it lighter to operate. The remarkable trail marks will be minimised as well.

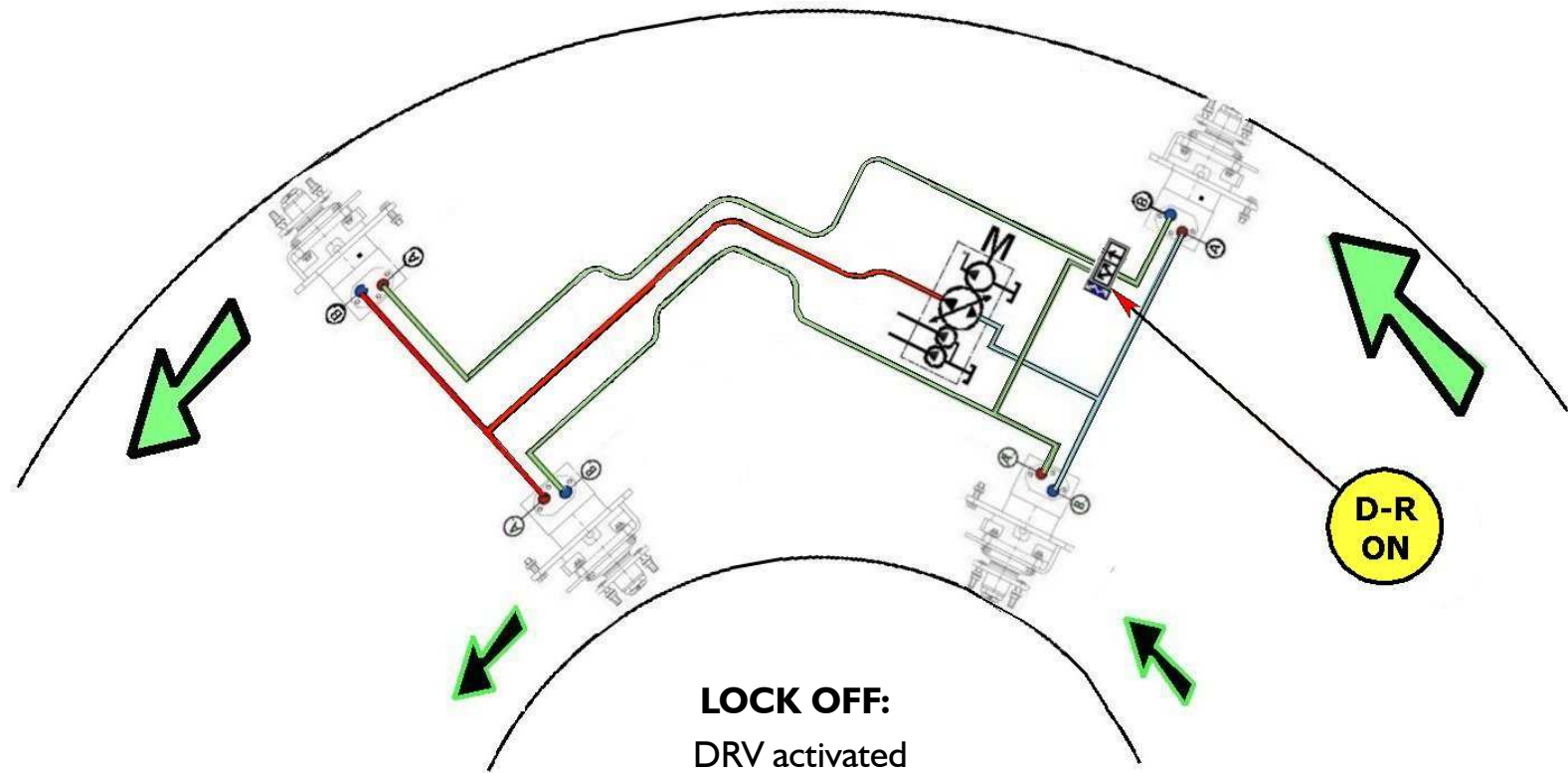


Example : An Avant 528, with 26x12-12 tyres. Drive the smallest possible circuit outer diameter about: 4,2 meters. The outer tyres require 6,7 turns and the inner ones 2,3 turns. The ratio of oil consumption is 1 : 3 between the inner and outer motors

Avant Drive Release Valve



Avant Drive Release Valve

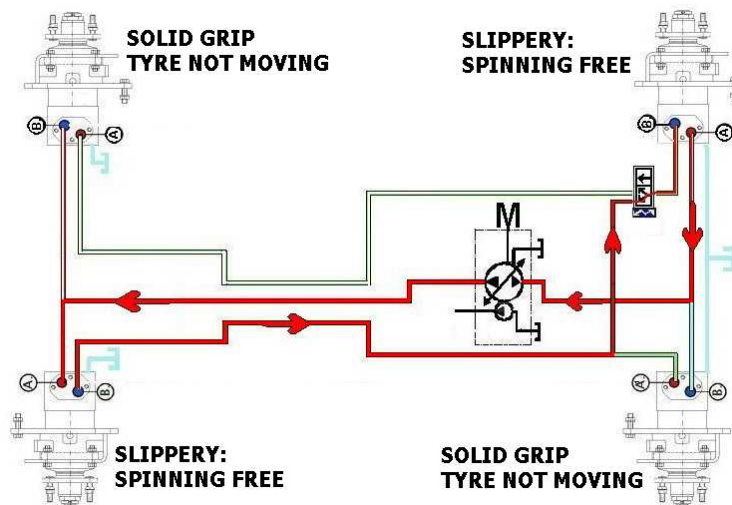


Avant Drive Release Valve

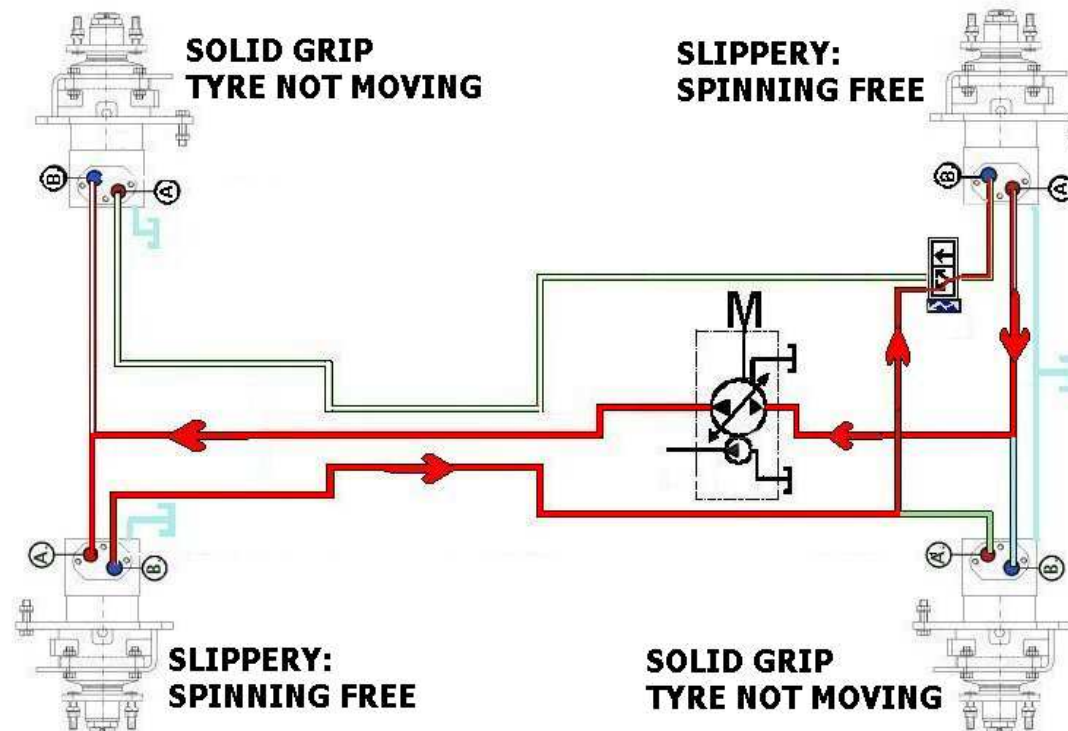
Using the drive release valve, may occasionally create a troublesome situation:

If, for some reason the drive release valve has been left active LOCK OFF, and one is operating the loader in a rough and difficult terrain, where the tyres at crossed opposite sides would remain hanging either on some hollow space or some slippery surface.

Then driving the loader may stop, and the crossed tyres may rotate free on high speed. Commonly the easiest way out, is to set the LOCK ON, or one could apply relief by pushing the loader with the telescopic boom.



Avant Drive Release Valve

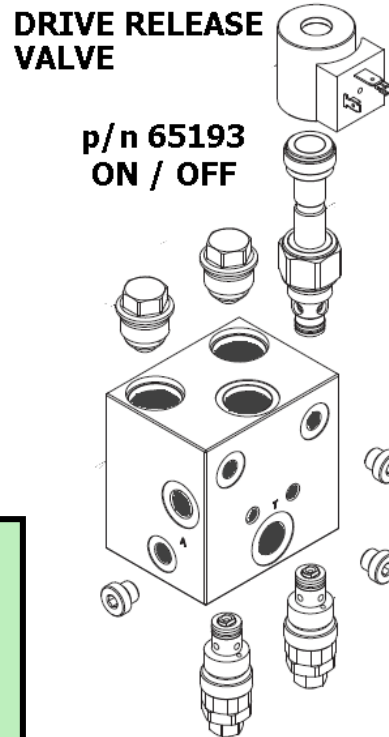


Avant Drive Release Valve

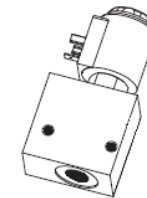
This valve has always been used with POCLAIN drive motors.

The valve includes two check valve cartridges, which efficiently protect the components of the hydraulic cavitation phenomenon.

Avant 200 series loaders have a manually operable flow control valve, a restrictor which can be fully closed to the LOCK ON position.



DRIVE RELEASE VALVE
p/n 64676
ON / OFF



The valve above is the traditionally used DRV in all earlier Avant loader models. This is still being used in the 400 series loaders, still having the M+S drive motors.



Avant Drive Release-, and the Anti Slip Valve

The anti slip valve either divides or collects the oil flow delivered by the variable drive line pump in even ratio, between the left and the right side drive motor pairs as precisely as possible. Anyhow it allows a slight volumetric flow difference in between the both sides. The speed and volume of the flow determines how precisely the flow divider valve works.

Traditionally the Anti Slip Valve has been seated in the front chassis, between the drive motors. As the Poclain drive motors were taken in use the arrangement is somewhat different. In the 700 series loaders the anti-slip valve was still situated in the front chassis, between the drive motors. But in the 500 and the 600 series loaders the position was changed different. The valve can be found from the front part of the rear chassis. It was seated behind the left front cover, when looking to the driving direction. In an other hand the POCLAIN drive release valve : p/n 65193, could be found from the front chassis between the front drive motors, correspondingly it was seated there in those loaders till the end of the year 2017.



Avant Drive Release-, and the Anti Slip Valve

Earlier the anti slip valve didn't have any special ON/OFF switch, but along the POCLAIN drive motors the valve now has a red color trigger switch, at the handle of the boom control joy-stick, which needs be used, whenever the anti-slip option will become necessary. It will be operated by pushing down the trigger switch and kept it pressed down, until the function will not be needed any more.

Optidrive became a standard hydraulic arrangement in Avant loaders, first in the 700-, e5 and e6 series loaders after the week 10/2017.

Rapidly after that it spread out to cover all the 500 and the 600 series loader models. In the beginning of 2018 it was the standard in all Avant loaders, excluding the 200 and 400 series, which still are using the conventional M+S orbital type driving motors.

Avant R-series also received the Optidrive equipment with the Poclain type drive motors in June of the year 2018.

Avant Drive Release-, and the Anti Slip Valve



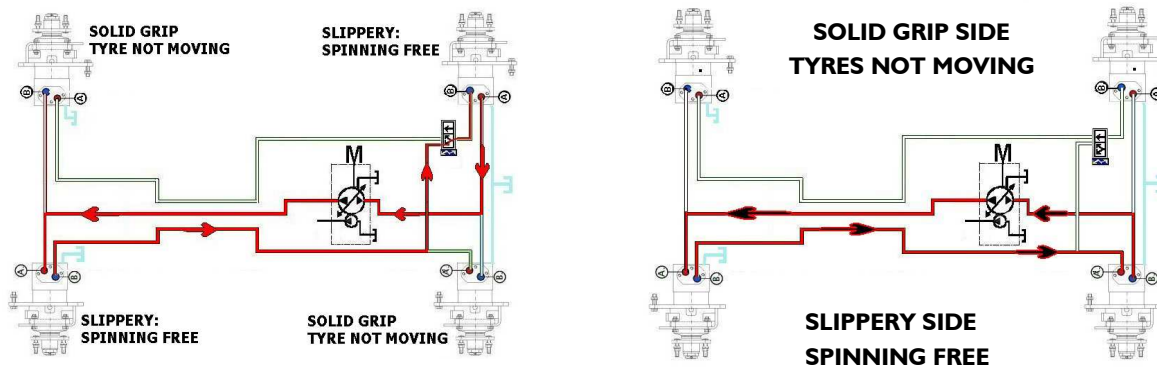
In Optidrive equipped loaders the hydraulic pump assembly has an integrated hydraulic block which carries the Anti-Slip valve.

In those loaders the drive release valve has been seated into another hydraulic block at the front chassis Fitted closely with the return oil filter assembly. The valve is the one at the right side when seeing it in front of the loader.



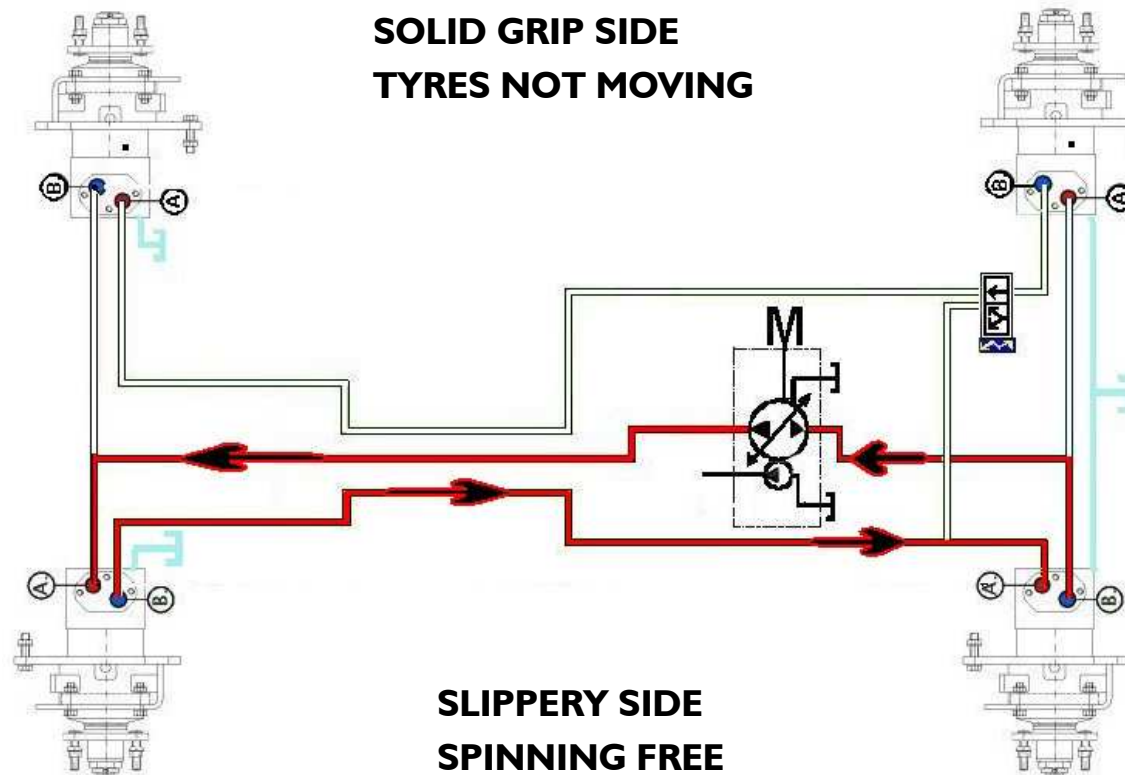
Avant Drive Release-, and the Anti Slip Valve

The anti slip valve prevents efficiently such situation to happen in which the tyre pair on the other side of the loader gets to a slippery surface and the other side remains on a solid grip, good friction surface, and the tyres on the slippery side begin to spin free. The correspondent other situation was the earlier mentioned, where the tyres at opposite, crossed corners remain on slippery area and begin to spin. In both cases the hydraulic fluid escapes throughout the driving circuit using the easiest way taking the minimum charge, leaving the loader immobile.



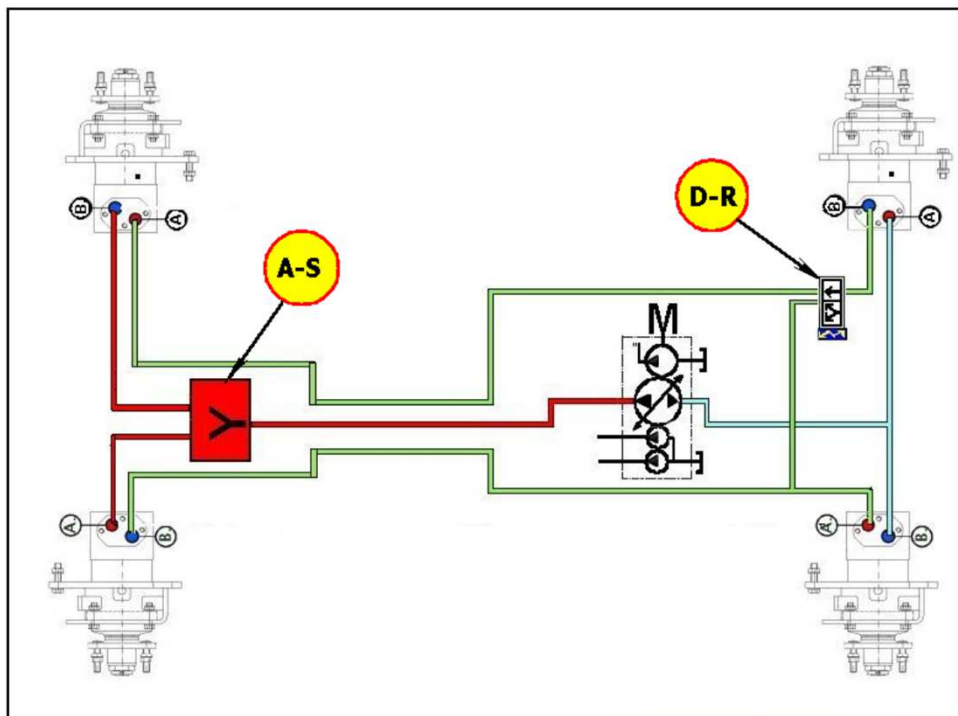
Avant Drive Release-, and the Anti Slip Valve

SITUATION WITHOUT THE ANTI SLIP VALVE



Avant Drive Release-, and the Anti Slip Valve

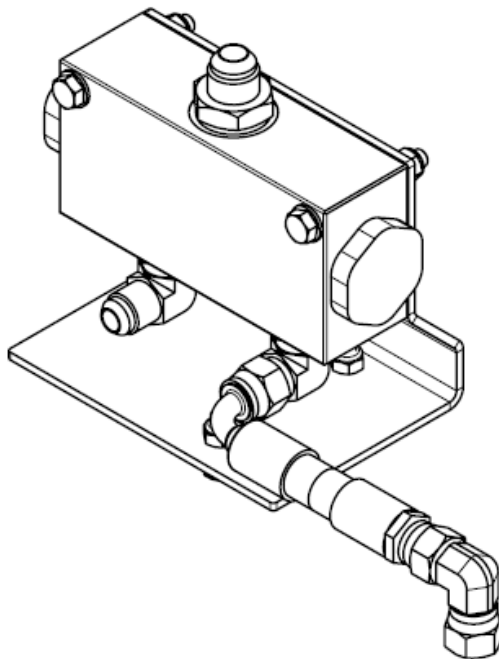
On a rough terrain areas one gets the best traction, when setting the engine RPM as high as possible, engages the LOCK ON position to the drive release valve, simultaneously gripping the anti slip, -trigger switch, holding it pressed down, when driving friskily.



AT THE IMAGE :

A-S = Anti Slip Valve,
D-R = Drive Release Valve,

Avant Drive Release-, and the Anti Slip Valve

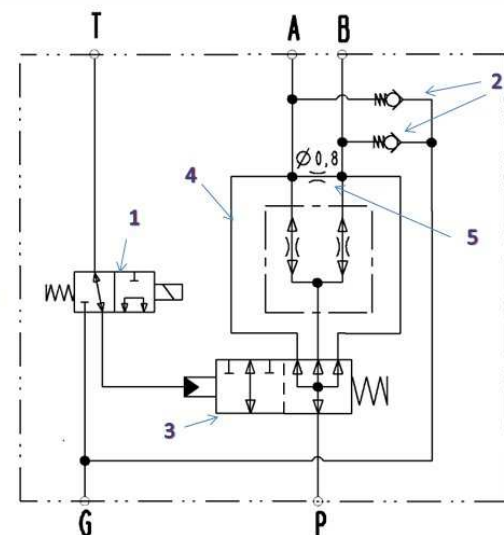
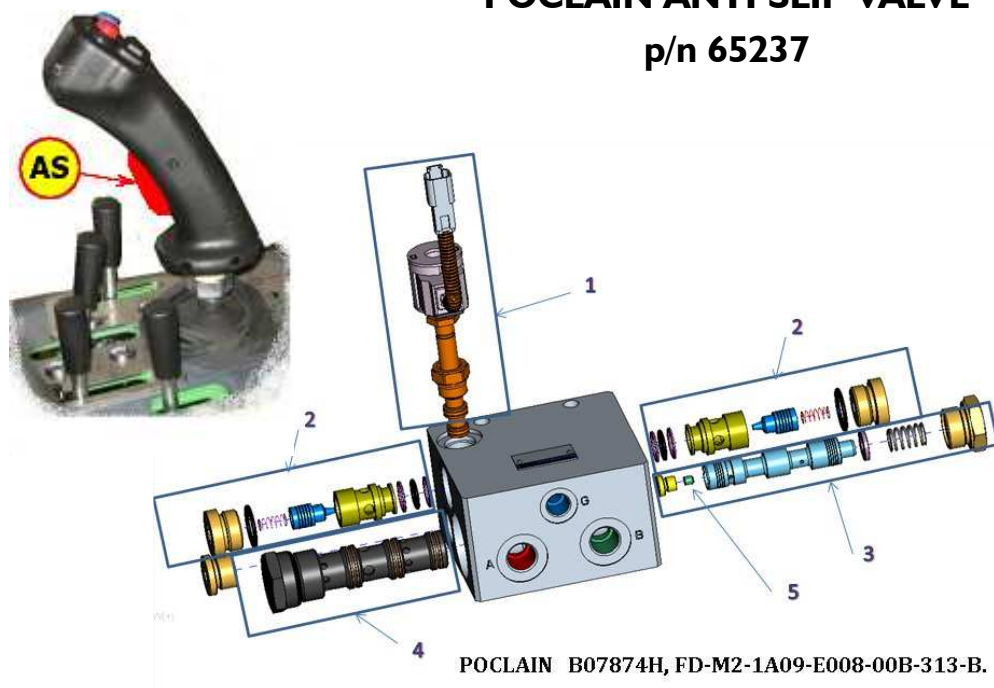


Anti slip valve: flow divider valve
p/n A411239 Avant 400-, 520 series
p/n A410193 Avant 200-, 528-, 600-, series
This was also used in Avant 745
series loaders before the POCLAIN
drive motors were taken in use
in the end of the year 2014 .

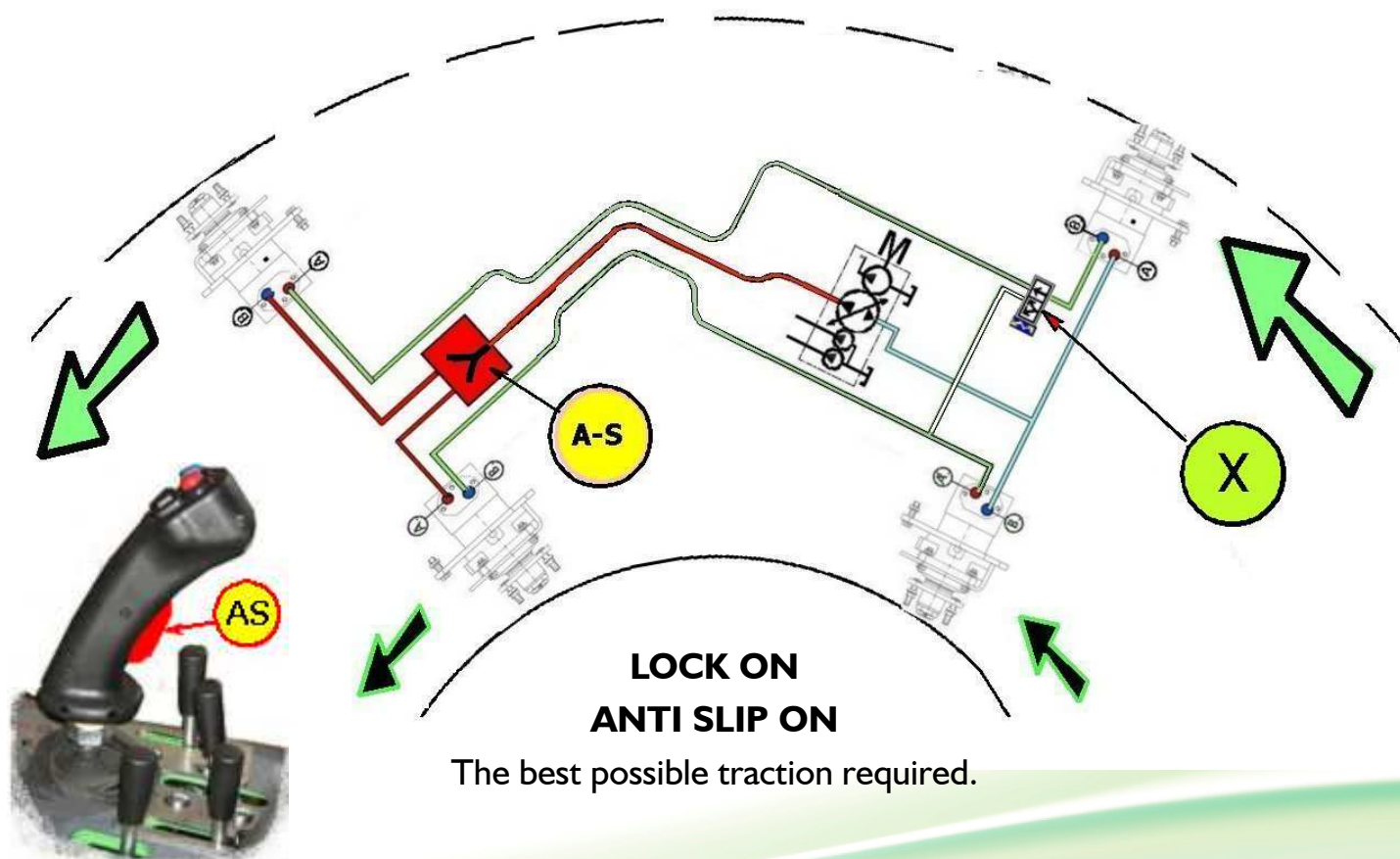
Avant Drive Release-, and the Anti Slip Valve

POCLAIN ANTI SLIP VALVE

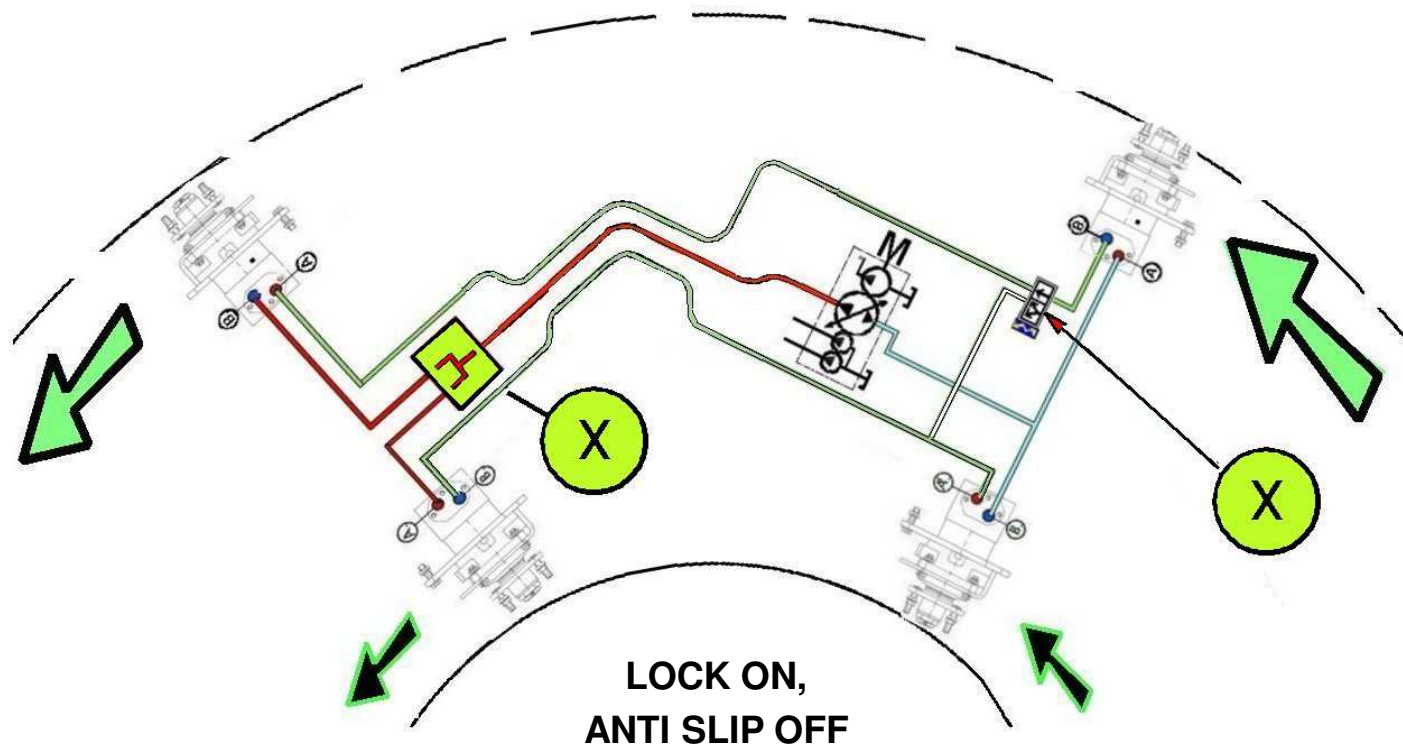
p/n 65237



Avant Drive Release-, and the Anti Slip Valve

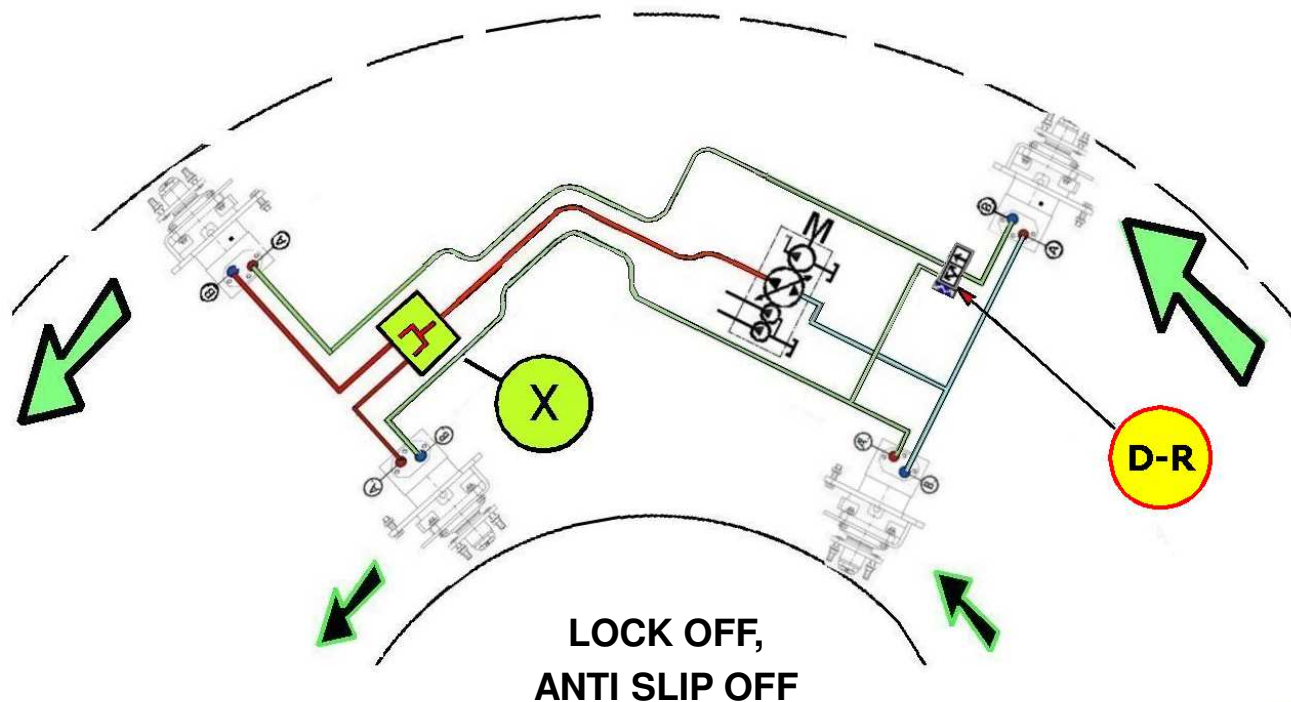


Avant Drive Release-, and the Anti Slip Valve



The normal driving situation without any special requirements.

Avant Drive Release-, and the Anti Slip Valve



Divng situation when driven on sensitive surface.

Leaving the track marks can be avoided and the loader remains light to be steered.